MDC AI & Robotics Lab Digital Twin & Multi-Method Simulation

Comprehensive Project Accomplishments, Architecture Overview, and Presentation Guide

👥 PROJECT TEAM & WORK BREAKDOWN STRUCTURE

Jorge A Mendoza — *Project Manager & AnyLogic Model Architect*
• DES process flows, ABM robot statecharts, 3D lab layout

Illia Shybanov — *Dashboard & Integration Specialist*
• Streamlit interactive application, live telemetry, and KPIs

Jordan Gonzalez — *Reinforcement Learning Engineer*
• Custom Gymnasium environment, Q-Learning & PPO dispatching

Ricardo Molina Gonzalez — *Simulation Experiments & Documentation Lead*
• 5 mandatory AnyLogic experiments, technical report, presentation deck

1. Executive Summary: What Has Been Accomplished

This project delivers a complete, multi-tiered digital twin and simulation system for the Miami Dade College AI & Robotics Lab. The system optimizes student work order dispatching, autonomous mobile robot (AMR) material delivery, and electric vehicle battery charging to eliminate chronic laboratory wait times and mid-corridor robot stalls.

System Component	Lead Contributor	Deliverable File / Location	Operational Status
AnyLogic 8 Multi-Method Model	Jorge A Mendoza	`anylogic_model/MDC_Robotics_Lab_Simulation.alp`	**Complete & Validated**
Supervisory Reinforcement Learning (PPO & Q-Learning)	Jordan Gonzalez	`rl_models/robotics_lab_env.py` , `train_ppo.py`	**67.3% Wait Reduction**
Interactive Streamlit Dashboard	Illia Shybanov	`dashboard/app.py` (Running at `http://localhost:8501`)	**Live & Operational**
3D WebGL Digital Twin & Drone Touchdown	Jorge & Ricardo	`digital_twin_app/` (Running at `http://localhost:3000`)	**Live & Verified**
Formal Academic Report (IEEE/MDC)	Ricardo Molina Gonzalez	`docs/Final_Project_Report_MDC.pdf` & `.html`	**Ready for Submission**
Oral Presentation Deck (16:9 Widescreen)	All Team Members	`docs/Presentation_Slides_MDC.pdf` & `.html`	**9 Slides with Timestamps**
Unified Canvas Archive	All Team Members	`MDC_Robotics_Lab_Group_Final_Submission.zip` (2.9 MB)	**Archived & Ready**

2. Detailed Technical Breakthroughs

🏗️ 1. AnyLogic Multi-Method Simulation (`.alp`)

- **Discrete-Event Flowchart:** Student request Poisson arrival → FIFO/Priority Order Queue → Seize AMR → Pick Delay (Station A) → Transit Delay → Release → Metric Sink.
- **Agent-Based Statecharts:** AMR class with continuous battery depletion (2%/min in transit) and inductive fast-charge (8%/min at Station D).
- **5 Configured Experiments:** Fleet Sizing Optimization, Compare Runs (FIFO vs. RL), Monte Carlo (100 runs), Sensitivity Analysis, and Stopwatch Calibration ($R^2 = 0.992$).

🧠 2. Reinforcement Learning Supervisory Control

- **67.3% Wait Time Reduction:** Average student waiting dropped from **9.8 minutes down to 3.2 minutes**.
- **Elimination of the "Charging Cascade":** Under simple threshold charging, all robots hit 20% simultaneously around midday. The PPO agent learned *opportunistic top-off charging* during demand lulls, eliminating midday fleet shutdowns.
- **Zero Battery Depletions:** Prevented robots from ever stalling mid-aisle with 0% battery.

🌐 3. 3D WebGL Digital Twin & Aerial Drone Integration

- **Heterogeneous Fleet:** Ground AMRs (AMR-01..03), Unitree G1 humanoid, and SkyWatcher drones (UAV-01..03).
- **Vertical Touchdown & Takeoff:** Aerial drones fly at 1.9m, descend vertically to ground level (0.22m) at Central Depot and building drop-off bays, release packages, and climb vertically before resuming cruise.
- **Live Package Drop-Off Counters:** Real-time delivery badges above each bay (e.g. `📍 Bay 1 Drop-Off [ 2 pkgs ]`) and in the sidebar buttons.
- **Bidirectional Live Bridge:** AnyLogic communicates directly with the 3D twin over `/api/anylogic/update`.

📊 4. Streamlit Analytics Dashboard (`http://localhost:8501`)

- Features all 4 team members' names, roles, and institutional affiliations in the sidebar.
- Interactive controls for Fleet Sizing (1–6 AMRs), Charging Pads (1–4), and Rush Hour demand surges.
- 5 Analytical Tabs: Operational KPIs, RL Policy Comparisons, AnyLogic Experiment Curves, 3D Telemetry, and Executive Summary.

3. Presentation Slide Deck & Speaker Schedule (10–12 Minutes)

Slide	Topic	Designated Speaker	Target Duration
Slide 1	Title Slide & Project Overview	**Jorge A Mendoza**	`0:00 - 0:45`
Slide 2	Problem Description & Laboratory Bottlenecks	**Jorge A Mendoza**	`0:45 - 2:00`
Slide 3	Multi-Method Methodology (Why DES + ABM?)	**Jorge A Mendoza**	`2:00 - 3:30`
Slide 4	AnyLogic 8 Implementation & Spatial Scale	**Jorge A Mendoza**	`3:30 - 5:00`
Slide 5	Supervisory Reinforcement Learning (PPO vs. Q-Learning)	**Jordan Gonzalez**	`5:00 - 6:45`
Slide 6	Live Streamlit Dashboard Walkthrough	**Illia Shybanov**	`6:45 - 8:15`
Slide 7	The 5 AnyLogic Simulation Experiments	**Ricardo Molina Gonzalez**	`8:15 - 9:45`
Slide 8	Non-Obvious Insights ("Fleet Sizing Trap" & "Charging Cascade")	**Ricardo Molina Gonzalez**	`9:45 - 11:00`
Slide 9	Conclusion, Deliverables Summary & Q&A	**All Team Members**	`11:00 - 12:00`

4. GitHub Repository & Submission Artifacts

All code, models, documentation, and simulation experiments are hosted and version-controlled at:
GitHub Repository: `https://github.com/AI4666/mdc-robotics-simulation-twin`

- `anylogic_model/MDC_Robotics_Lab_Simulation.alp` — Primary AnyLogic 8 Project File
- `dashboard/app.py` — Streamlit Web Dashboard
- `rl_models/train_ppo.py` & `train_qlearning.py` — Reinforcement Learning Training Code
- `docs/Final_Project_Report_MDC.pdf` — Formal Academic Paper (PDF)
- `docs/Presentation_Slides_MDC.pdf` — 9-Slide Widescreen Presentation (PDF)

- `MDC_Robotics_Lab_Group_Final_Submission.zip` — Final Unified Canvas Submission Package